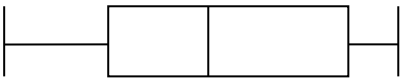
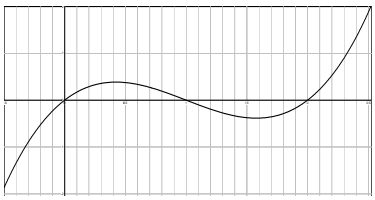
Question	Answer	Marks	Partial Marks
7(c)(ii)	$(2x-9)(5x+2)$ or $\dfrac{-(-41)\pm\sqrt{(-41)^2-4(10)(-18)}}{2(10)}$	**M2**	**B1** for $(ax+b)(cx+d)$ with $ac = 10$ and $bd = -18$ or $ad + bc = -41$ or $\sqrt{(-41)^2-4(10)(-18)}$ or $\dfrac{-(-41)+\sqrt{q}}{2(10)}$ oe or $\dfrac{-(-41)-\sqrt{q}}{2(10)}$ oe or both or **M1** for $\left(x-\dfrac{41}{20}\right)^2-\dfrac{18}{10}-\left(\dfrac{41}{20}\right)^2=0$ or better
	10	**A2**	**A1** for $[x =]\ \dfrac{9}{2}$ oe or **M1** for $2 \times$ *their* positive root + 1
8(a)(i)	$\dfrac{60}{360}\times 600$ oe	**1**	
8(a)(ii)	45	**2**	**M1** for $\dfrac{27}{360}\times 600$ oe
8(a)(iii)	Correct straight line on the pie chart	**2**	**B1** for 75
8(b)	Correct diagram  0.6 3.4 5.2 7.5 8.7	**3**	**B1** for any three of 0.6, 3.4, 5.2, 7.5, 8.7 correctly placed **B1** for 7.5 and 8.7 seen
8(c)(i)	5	**1**	
8(c)(ii)	2	**1**	
8(c)(iii)	3	**1**	
8(d)	39.2	**4**	**M1** for mid-values soi **M1** for Σfx with x in correct interval including boundaries **M1 dep** for $\dfrac{\Sigma fx}{50}$ dep on second M1
9(a)	(0, 0), (1, 0), (2, 0)	**2**	**B1** for any two correct If 0 scored, SC1 for all three x values clearly identified

Question	Answer	Marks	Partial Marks
9(b)	$x\left(x^2-x-2x+2\right)$ or $\left(x^2-x\right)(x-2)$ or $(x-1)\left(x^2-2x\right)$ leading to x^3-3x^2+2x with no errors or omissions	**2**	**B1** for $x\left(x^2-x-2x+2\right)$ or $\left(x^2-x\right)(x-2)$ or $(x-1)\left(x^2-2x\right)$
9(c)	$3x^2-6x+2$	**B2**	**B1** for 2 correct terms
	their $\frac{dy}{dx}=0$	**M1**	
	their $\frac{-(-6)\pm\sqrt{(-6)^2-4(3)(2)}}{2(3)}$	**M2**	**M1** for $\sqrt{(-6)^2-4(3)(2)}$ or for $p=-(-6)$ and $r=2(3)$ if in form $\frac{p\pm\sqrt{q}}{r}$
	(0.4, 0.4) (1.6, –0.4)	**B3**	**B2** for 0.4 or 0.42... **and** 1.6 or 1.57 to 1.58 or for one correct pair of coordinates or **B1** for 0.4 or 0.42... **or** 1.6 or 1.57 to 1.58 If 0 scored **SC1** for $1+\sqrt{\frac{1}{3}}$ and $1-\sqrt{\frac{1}{3}}$ or better or for one correct pair of coordinates in any form
9(d)	Correct sketch 	**2**	**FT** *their* **(c)** but must be cubic i.e. correct shape cubic through origin and max and min in correct quadrants **B1** for cubic shape sketch
10(a)(i)	1	**1**	
10(a)(ii)	$\frac{1}{4}$ oe **nfww**	**2**	**M1** for $\frac{2}{4}\times\frac{2}{4}$ oe
10(a)(iii)	7	**2**	**M1** for trials with $\left(\frac{3}{4}\right)^k\times\frac{1}{4}$ soi
10(b)(i)	0.72 oe	**2**	**M1** for 0.9×0.8
10(b)(ii)	0.26 oe	**3**	**M2** for $0.9\times0.2+0.1\times0.8$ or 1 – *their* **(b)(i)** – 0.1×0.2 or **M1** for 0.9×0.2 or 0.1×0.8 or 1 – *their* **(b)(i)** or $1-0.1\times0.2$
11(a)(i)	64	**1**	